ENBC332: Homework #1

Due: Monday, 09/15/25 by noon, beginning of the class

Submit your homework to ELMS, including R code and dataset

Overall points: 50

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: “problem1_fullname.R”. Start with the problem number and followed by your full name, e.g. “**problem1_jimmyazarnoosh**”.

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

R Programming

Problem 1. Write a R code to create three vectors **numeric data**, **character data** and **logical data**. Vectors' length between 5 to 10 is sufficient. Display the content of the vectors and their **type**. (10 Points)

Problem 2. Write a R code to create a vector with numbers from 10 to 50 and find the mean, median, and sum of these numbers. Get the cumulative sum, vector y, and plot it. (10 Points)

Problem 3. Write a R code to create a vector which contains 5 random integer values between 0 and 100. (10 Points)

Problem 4. Create an Excel file with a dataset that contains only **TWO columns** with numbers. In the first row, assign names to each column. You can find data sets online (Kaggle) or any other websites. Please add the reference link of the dataset you found online. Write a R code to import this dataset to the code using readxl library and then plot it. Get the mean, median, and sum for each column.

Please include an explanation of the dataset you used, along with a discussion of the relationship between these two columns. (20 Points)